EF3320 Assignment

Semester A, 2026/27

There are two versions of the individual assignment: **Core Version** and **Challenge Version**. Please choose one version only.

Students who complete the Challenge Version may earn up to 30% additional points based on the total assignment marks.

In this assignment, you are also required to use an AI tool as a learning assistant. AI should not replace your own analysis. Instead, use AI to check your solution, troubleshoot your work, or critique your final portfolio.

The deadline for submitting the assignment to Canvas is 11:59 p.m., 1 November 2026, Hong Kong time.

Core Version

Suppose that you are a fund manager managing a portfolio worth $100 million. You have adopted a fixed investment policy of 60% stocks as follows: 30% to large-cap stock portfolios (LRGSTK), 20% to medium-cap stock portfolios (MEDSTK), and 10% to small-cap stock portfolios (SMLSTK). The remaining portfolio is allocated 30% to Corporate Bonds (CBOND) and 10% to Government Bonds (GBOND).

In the spreadsheet Data for Assignment, you will find monthly returns for these five assets. The spreadsheet also provides returns on three additional asset classes: Real Estate (Equity REITs), International Stocks (EAFE), and Precious Metals (PM). In addition, it provides monthly returns on the value-weighted market portfolio (VWRETD), the S&P 500 Index (SPRET), and one-month U.S. Treasury bills (TBILL).

Please use the provided data to complete the analysis in Excel. Upload your completed Excel file to Canvas, including all detailed calculations and work. Highlight your final answers where appropriate. **Screenshots of Excel files will not be accepted as valid submissions.**

1. Provide performance statistics, including the annualized mean returns, annualized standard deviation, and Sharpe ratio for the following three funds over the full sample period:
    a. the fund under your management, based on the fixed investment policy described above;
    b. the value-weighted market portfolio (VWRETD);
    c. the S&P 500 Index (SPRET).

2. Plot the portfolio frontier using only the five risky assets currently included in your portfolio: LRGSTK, MEDSTK, SMLSTK, CBOND, and GBOND. Assume that a risk-free rate is available. Identify the optimal risky portfolio, also known as the tangency portfolio, on the frontier. Report the weight of each asset in the tangency portfolio, as well as the portfolio's expected return and standard deviation.

3. Is the current risky-asset portfolio efficient, meaning that it lies on the efficient frontier? If it is not efficient, calculate the investment proportions in the five risky assets required to construct an efficient risky portfolio that delivers the same expected return as the current risky-asset portfolio.

4. Assume that you can invest in the risk-free asset. Suggest an efficient portfolio allocation that achieves your client's objective of a **floor expected return of 1% per month**. Also calculate the expected standard deviation of this portfolio. Remember to annualize the floor rate of return when reporting annualized results.

5. You want to further diversify your portfolio holdings by including Real Estate, represented by Equity REITs; International Stocks, represented by the MSCI EAFE index, which includes stocks from Europe, Asia, and the Far East; and Precious Metals (PM). Plot the minimum variance frontier using all eight risky assets and compare it with the frontier using the original five risky assets.

6. Verify your answers using an AI tool of your choice. In no more than 200 words, list two things the AI helped you understand or fix, and identify the AI tool used. Please type your response directly in the Excel file.

Challenge Version

Suppose that you are a fund manager managing a portfolio worth $100 million. Instead of using the pre-provided asset return data, you will construct your own dataset using WRDS. You must select **6 risky assets** from WRDS and use their historical return data over the period **January 2020 to December 2024** to conduct a mean-variance optimization analysis. Your six assets should satisfy the following requirements:

- They must be U.S.-listed common stocks.
- They must have complete monthly return data (60 monthly observations) over your chosen sample period.
- They should represent at least **3 different industries**.

Use the one-month U.S. Treasury bill rate as the risk-free rate. You may obtain the risk-free rate from WRDS or another credible source (such as *Fama and French Data Library*), but you must clearly document the source.

Please use the data to complete the analysis in Excel. Upload your completed Excel file to Canvas, including all detailed calculations and work. If you choose other tools, such as Python, please also upload the relevant files, such as your Python code. Highlight your final answers where appropriate. **Screenshots of Excel files will not be accepted as valid submissions.**

1. Download monthly return data for your selected six assets from WRDS. In your file, clearly report ticker symbol and monthly returns.

2. Provide performance statistics, including the annualized mean returns, annualized standard deviation, and Sharpe ratio for your six assets. Use the one-month Treasury bill rate as the risk-free rate when calculating the Sharpe ratio.

3. Assume that your current investment policy is to allocate the $100 million equally across the six risky assets. Calculate annualized expected portfolio return, annualized portfolio standard deviation and Sharpe ratio.

4. Using your six selected assets, plot the minimum variance frontier in Excel. Assume that investors can also invest in the risk-free asset. Identify the optimal risky (tangency) portfolio on the frontier. Report the weight of each asset in the tangency portfolio, as well as the portfolio's expected return and standard deviation.

5. Determine whether the equally weighted portfolio is an efficient portfolio. If it is not efficient, calculate the investment proportions in the six risky assets required to construct an efficient portfolio that delivers the **same expected return** as the equally weighted portfolio.

6. Assume that you can invest in the risk-free asset. Suggest an efficient portfolio allocation that achieves your client's objective of a **floor expected return of 1% per month**. Also calculate the expected standard deviation of this portfolio. Remember to annualize the floor rate of return when reporting annualized results.

7. Verify your answers using an AI tool of your choice. Ask the AI tool to critique your portfolio from an investment perspective. In **no more than 200 words**, list two things the AI helped you understand or fix, and identify the AI tool used.

Data sources

Fama and French Data Library

The website provides data on the risk-free rate, sorted portfolio returns and risk factors (U.S.), including the market factor, Small-Minus-Big (SMB) size factor, High-Minus-Low (HML) value factor, momentum factor, investment factor, profitability factor, and others.

Wharton Research Data Services (WRDS)

WRDS is a platform that provides comprehensive data on firm fundamentals and stock markets. Although its data coverage is global, the data quality is better for firms listed on U.S. stock exchanges (NYSE, AMEX, and NASDAQ). To get access,

- Register with your CityU email address using the link below: https://wrds-www.wharton.upenn.edu/register/cityuhk/?user_type=class-student
- You will also need to provide the class code for EF3320, which is **7ff27dca** (for sections C01 and CA1) or **fa871ac8** (for sections C03 and CA3)
- After registration and login, you will find a wide range of data products provided by various data vendors. Stock market data are primarily maintained by CRSP.